Data Cleaning and Quality Report

SIMULATED DATA NOTICE: all results in this document are calculated from a synthetic dataset created to test the research pipeline. They are not findings about real students, and the dataset contains no genuine questionnaire responses.

1. Dataset

File: student_academic_performance_synthetic_1000.csv. Rows: 1000. Columns: 17. Data source flag: SYNTHETIC. Simulated timestamps span 2026-09-14 06:23:18 to 2026-10-25 21:35:59 and are **not** actual submission times.

2. Validation results

Duplicate student identifiers: 0. Fully duplicated response rows (ignoring identifier): 0. Rows whose category disagrees with the documented grading thresholds: 0. Total missing cells: 60 across 59 rows.

Variable	Allowed range	Observed min	Observed max	Missing (n)	Missing (%)	Out of range (n)
study_hours_weekly	0-80	0.0	33.0	6	0.6	0
attendance_percentage	0-100	30.0	100.0	11	1.1	0
assignment_completion_percentage	0-100	15.0	100.0	0	0.0	0
sleep_hours	2-14	3.5	10.0	14	1.4	0
number_of_courses	1-12	3.0	10.0	0	0.0	0
social_media_hours_daily	0-16	0.0	8.2	13	1.3	0
previous_performance	0-100	25.6	85.2	16	1.6	0
current_performance	0-100	23.3	88.5	0	0.0	0
year_of_study	1-5	1.0	5.0	0	0.0	0

3. Descriptive statistics and outliers

Outliers were identified with the 1.5 x IQR rule and **inspected, not removed**: values stay inside plausible ranges, and extreme but possible answers (for example long study hours) are legitimate variation.

Variable	Mean	SD	Min	Median	Max	Missing	IQR outliers
study_hours_weekly	12.16	7.18	0.0	12.0	33.0	6	1
attendance_percentage	75.66	13.15	30.0	76.0	100.0	11	4
assignment_completion_percentage	77.58	14.74	15.0	78.0	100.0	0	5
sleep_hours	6.6	1.13	3.5	6.6	10.0	14	11
number_of_courses	6.27	1.35	3.0	6.0	10.0	0	0
social_media_hours_daily	3.59	1.61	0.0	3.6	8.2	13	2
previous_performance	56.42	10.27	25.6	56.2	85.2	16	3
current_performance	56.37	10.09	23.3	56.7	88.5	0	9

4. Cleaning decisions

Issue	Decision	Reason
Missing values (about 1% per affected column)	Keep rows. Impute inside the modelling pipeline: median for numeric columns, most frequent for categorical columns, fitted on training data only.	Avoids discarding rows (small samples) and avoids test-set information shaping the model.
Out-of-range values	None found; range rules retained for real data (percentages 0-100, sleep 2-14 h, courses 1-12, etc.).	Validation must be re-run on genuine responses.
Duplicates	None found. With real data check repeated submissions (identical rows within minutes).	Anonymous forms cannot rule out repeats; flag, do not auto-delete.

Issue	Decision	Reason
Outliers	Kept; model scaling uses standardisation (LR), none needed for Random Forest.	Removal would shift the sample toward unrealistically tidy students.
Target definition	Derived only from current_performance with fixed thresholds (High >=60, Medium 50-59.9, Low <50).	Prevents the target from being reconstructed from predictors.

5. Machine-learning readiness

Topic	Recommendation
Features	Study hours, attendance, assignment completion, sleep, internet access, study location, number of courses, social media hours, previous performance (9 predictors).
Excluded	student_id, university, programme, year_of_study, timestamp, data_source (identifiers/administrative, no scientific justification as predictors) and current_performance (the target is derived from it: leakage).
Missing values	SimpleImputer (median / most frequent) inside a Pipeline, fitted on the training partition only.
Encoding	One-hot encoding for internet_access and study_location; unseen categories ignored.
Scaling	StandardScaler for numeric predictors (needed for regularised Logistic Regression, harmless for Random Forest).
Split	Stratified 80:20 split (train 800, test 200), seed 42; 5-fold stratified CV on the training part for model selection; test set used once. With the planned 50-150 real students the test set will hold only about 10-30 people, so use fewer folds and report uncertainty.
Leakage prevention	Fit every transformer inside the pipeline; never fit on the full dataset; keep current_performance and its derivatives out of X.
Class imbalance	Counts: Low 264, Medium 381, High 355. Majority share 38.1%, largest-to-smallest ratio 1.44. Mild imbalance: use stratification and macro-F1; consider class_weight only if a class becomes small in real data.

6. Is previous performance redundant with the target?

Pearson correlation between previous and current performance is 0.643 (Spearman 0.619, n=984). Test macro-F1 for Logistic Regression / Random Forest: all nine predictors 0.556 / 0.549; without previous performance 0.454 / 0.462; previous performance alone 0.570 / 0.561. Previous performance carries a large share of the usable signal here. It refers to an earlier period, so it is legitimate history rather than direct leakage, but it should be reported both with and without it. These are single-split results on 200 test rows, so differences of about 0.01 are within noise. The pattern comes from the simulator's design and says nothing about real students.